4

Orbits, or how things stay up

A satellite is not floating. It is falling, all the time, and missing the ground.

BY THE END OF THIS CHAPTER YOU SHOULD BE ABLE TO

- explain why an object in orbit is in free fall, using Newton's cannon as a thought experiment;
- calculate the speed and period of a circular orbit from its radius;
- describe an elliptical orbit with its semi-major axis, eccentricity, periapsis and apoapsis;
- use Kepler's third law to compare the orbits of planets and satellites;
- explain the difference between orbital speed and escape speed.

4.1 Falling around the Earth

Key idea. An orbit is a fall that never reaches the ground because the ground curves away as fast as the object drops.

Astronauts on the International Space Station float around the cabin, and photographs of them are often described as showing "zero gravity". That description is wrong. At the station's height, about 400 km above the surface, the pull of the Earth is still about 89% of what it is on the ground. If you could build a tower 400 km tall and stand on top of it, you would feel almost exactly as heavy as you do now.

So why do astronauts float? Because the station and everyone inside it are falling together. A person in a falling lift feels weightless for the same reason: the floor drops away at the same rate as their feet, so it pushes on

them with no force at all. The difference is that the lift hits the bottom of the shaft after a second or two, while the station keeps falling for years.

The station never hits the ground because it is also moving sideways, very fast: about 7.7 kilometres every second. In the time it takes to fall a few metres, it travels so far around the planet that the surface beneath it has curved away by the same few metres. It falls, and it misses.

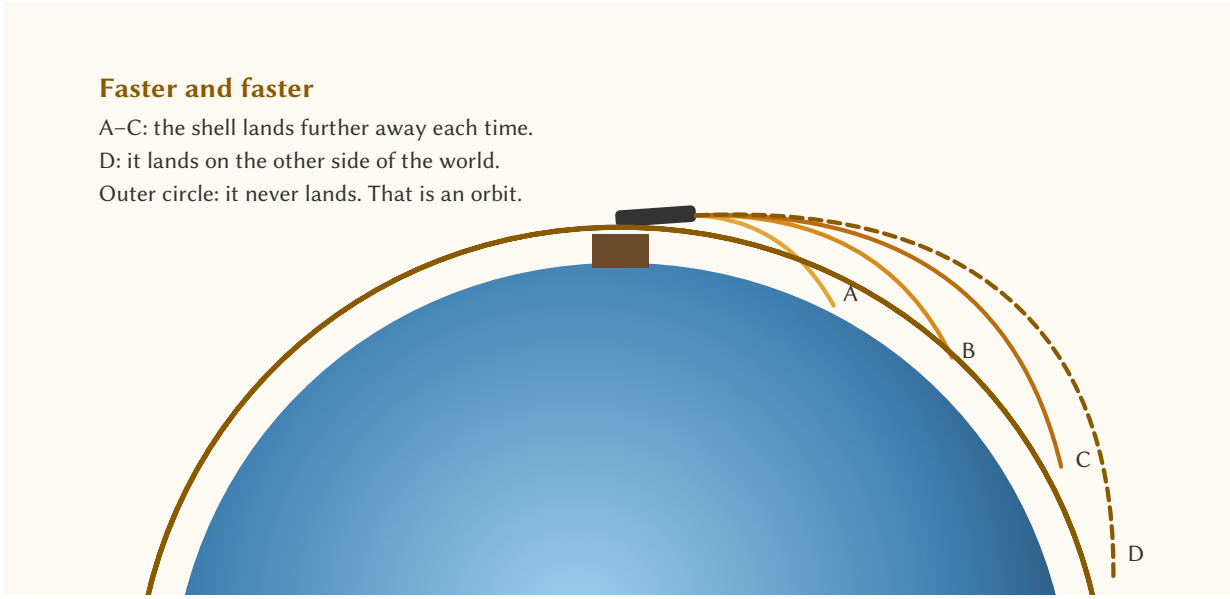


Figure 4.1 Newton's cannon. A cannon on an impossibly tall mountain fires shells horizontally at greater and greater speeds. Each shell falls, but the faster ones travel further before they reach the ground, until one is fast enough to fall all the way round. Newton drew a similar picture in his Treatise of the System of the World, published in 1728.

Isaac Newton described this as a thought experiment, illustrated in Figure 4.1. Imagine a mountain so high that it rises above the air, so there is no drag. A cannon on its summit fires a shell horizontally. The shell curves down and lands some distance away. Fire it faster, and it lands further away. Because the Earth is round, there is a speed at which the shell falls towards the ground exactly as fast as the ground curves away beneath it.[1] That shell never lands. It comes back round to the mountain from behind, and, if nobody moves the cannon, it carries on circling forever.

1. Newton's own diagram shows the mountain as a small bump on a large circle. The mountain would need to be well over a hundred kilometres high to rise above most of the atmosphere, which is why the experiment stays in the imagination.

4.2 Circular orbits

For the simplest orbit, a circle, we can work out the speed needed. An object moving in a circle of radius r at speed v needs a force towards the centre of mv^2/r. In orbit, gravity provides exactly that force, and nothing else does. Newton's law of gravitation gives the size of the gravitational force between the Earth, of mass M, and the satellite, of mass m:

$$F = GMm / r^2 \tag{4.1}$$

Setting the two equal, the satellite's mass cancels out. That is a remarkable result: a spanner dropped by an astronaut and the station itself follow the same orbit, which is why tools that drift away during a spacewalk stay close to the station for a while before slowly separating.

$$v = \sqrt{(GM / r)} \tag{4.2}$$

The product GM for the Earth is known very precisely from the motion of satellites: 3.986×10^{14} m^3 s^{-2}. You do not need to know G and M separately to calculate an orbit, which is fortunate, because G is one of the least precisely measured constants in physics.

Notation. We use r for the distance from the centre of the Earth, not the height above the surface. For the station, r ≈ 6371 km + 400 km = 6771 km.

WORKED EXAMPLE: THE SPACE STATION

The station orbits about 400 km above the surface, so $r = 6.771 \times 10^6$ m. Equation (4.2) gives

$$v = \sqrt{(3.986 \times 10^{14} / 6.771 \times 10^{6})} \approx 7670 \text{ m s}^{-1}$$

One trip round the circle is $2\pi r \approx 4.25 \times 10^7$ m, so the period is

$$T = 2\pi r / v \approx 5540 \text{ s} \approx 92 \text{ minutes.}$$

The crew see about sixteen sunrises and sixteen sunsets every day.

Equation (4.2) says that higher orbits are slower, as Figure 4.2 shows. This surprises most people the first time they meet it. A satellite 20,000 km up moves at under 4 km s^{-1}, about half the station's speed, and because it also has much further to go round, its period is nearly twelve hours. The satellites of the Global Positioning System orbit at about that height.

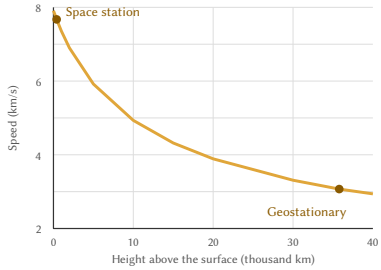


Figure 4.2 The speed of a circular orbit falls as the orbit gets higher, from Equation (4.2).

Go higher still and the period stretches to a full day. At a radius of 42,164 km, a satellite above the equator goes round once in 23 hours and 56 minutes, the time the Earth takes to turn once relative to the stars.[2] Seen from the ground, such a satellite hangs still in the sky, so a dish pointed at it never needs to move. This is a *geostationary* orbit, and it is crowded with weather and communication satellites.

Table 4.1 Common Earth orbits.

Orbit	Height	Period
Low (LEO)	160–2000 km	88–127 min
GPS (MEO)	20,200 km	11 h 58 min
Geostationary	35,786 km	23 h 56 min
The Moon	≈ 384,000 km	27.3 days

Table 4.1 lists some common orbits. Notice how strongly the period depends on the height. Low orbits are cheap to reach and close enough for sharp photographs of the ground, but a satellite there is only above any one place for a few minutes at a time. Higher orbits see more of the Earth at once, but the signals take longer to travel and the satellites cost more to launch. Engineers choose an orbit the way an architect chooses a site: by balancing what the mission needs against what it can afford.

The Moon itself is a satellite in a nearly circular orbit, at about sixty times the radius of the Earth. Newton's great insight was that the force making an apple fall and the force holding the Moon in its orbit are the

2. That is a sidereal day. The 24-hour solar day is about four minutes longer, because the Earth also moves along its orbit round the Sun during the day and has to turn a little further to bring the Sun back to the same place in the sky.

same force, weakened at the Moon's distance by the square of sixty. He checked that idea against the Moon's period and found that it agreed "pretty nearly", in his own words.

4.3 Ellipses

Most real orbits are not circles but ellipses, as Johannes Kepler discovered in 1609 from Tycho Brahe's careful observations of Mars. An ellipse is a stretched circle with two special points inside it called foci. The body being orbited sits at one focus, as Figure 4.3 shows, and the other focus is empty.

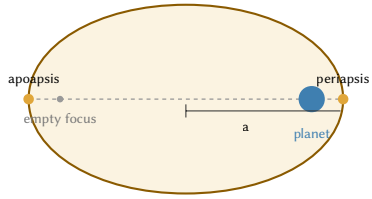


Figure 4.3 An elliptical orbit. The planet sits at one focus; the other focus is empty.

The size of an ellipse is given by its semi-major axis a, half of its longest diameter. Its shape is given by its eccentricity e, a number between 0 and 1. A circle has $e = 0$; the larger e, the more stretched the ellipse. The Earth's orbit round the Sun has $e \approx 0.017$, so close to a circle that you could not tell the difference by eye. Halley's comet has $e \approx 0.97$, and spends most of its 76-year orbit far out beyond Neptune.

Words. Periapsis and apoapsis are the general terms. For orbits round the Earth they are often called perigee and apogee; round the Sun, perihelion and aphelion.

The point of closest approach is the periapsis and the furthest point is the apoapsis. A body in an elliptical orbit moves fastest at periapsis and slowest at apoapsis. Kepler expressed this as his second law: a line from the Sun to a planet sweeps out equal areas in equal times. Today we see it as a consequence of conservation of angular momentum, which you met in Chapter 3.

Changing orbit

To move a spacecraft from a low circular orbit to a higher one, mission planners use a *transfer orbit*: an ellipse whose periapsis touches the low orbit and whose apoapsis touches the high one. A short burn of the engine at the low orbit stretches the circle into the ellipse; half an orbit later, at apoapsis, a second burn rounds the ellipse out into the higher circle. This two-burn manoeuvre is called a Hohmann transfer, after the German engineer Walter Hohmann, who described it in 1925, decades before anyone could launch a satellite.

TRY IT YOURSELF

1. Tie a weight to a length of string and swing it round in a horizontal circle. Feel how the pull you must supply grows as you swing faster. That pull plays the part of gravity.
2. Pin a loop of string round two drawing pins on a board and trace a pencil round the inside of the loop. You have drawn an ellipse with the pins at its foci. Move the pins apart and the ellipse becomes more eccentric.

4.4 Kepler's third law

Kepler's third law, published in 1619, connects the size of an orbit to its period. For every planet going round the Sun, the square of the period is proportional to the cube of the semi-major axis:

$$T^2 / a^3 = 4\pi^2 / GM = \text{constant} \tag{4.3}$$

If we measure T in years and a in astronomical units (the average distance from the Earth to the Sun), the constant is exactly 1 for every planet, because it is 1 for the Earth by definition. Table 4.2 shows how well the planets obey it.

Table 4.2 Kepler's third law for six planets. Semi-major axes in astronomical units, periods in years.

Planet	Semi-major axis a	Period T	T^2	a^3	T^2 / a^3
Mercury	0.387	0.241	0.0581	0.0580	1.00
Venus	0.723	0.615	0.378	0.378	1.00
Earth	1.000	1.000	1.000	1.000	1.00
Mars	1.524	1.881	3.54	3.54	1.00
Jupiter	5.203	11.86	140.7	140.8	1.00
Saturn	9.537	29.46	867.9	867.4	1.00

Kepler found this law by trial and error, working through years of calculations by hand. Newton later showed that it follows directly from his law of gravitation, and in doing so gave the constant a meaning: it depends only on the mass of the body at the focus. That turned Kepler's law into a scale. Measure the period and size of any orbit, and you can

weigh whatever is at the centre, whether it is a planet with moons, a star with planets, or the black hole at the centre of our galaxy.

The same law works for the Moon and for artificial satellites, with the Earth's mass in place of the Sun's. The Apollo spacecraft that orbited the Moon, like the one whose crew took the photograph in Figure 4.4, used it with the Moon's mass: in a low orbit about 110 km above the lunar surface they went round once every two hours.



Figure 4.4 The Earth seen from lunar orbit by the Apollo 8 crew in 1968 (NASA, public domain).

4.5 Escaping

What happens if the cannon on Newton's mountain fires even faster than the speed for a circular orbit? The shell still goes round, but its path stretches into an ellipse that carries it further and further out. At a speed $\sqrt{2} \approx 1.41$ times the circular speed, the ellipse never closes: the shell has enough energy to escape from the Earth's gravity altogether and never come back. That speed is called the escape speed.

Key idea. Escape speed is $\sqrt{2}$ times the circular orbital speed at the same distance.

At the Earth's surface, ignoring air, the circular orbital speed is about 7.9 km s^{-1} and the escape speed is about 11.2 km s^{-1}. Notice that the escape speed does not depend on the direction of travel, as long as the path does not run into the planet: it is a statement about energy, not about aim.[3]

The worked example about the space station on page 3 used circular orbits. The questions below ask you to apply the same ideas to ellipses and escape.

3. Real rockets do care about direction, because the Earth's rotation gives anything launched eastwards from near the equator a free start of up to 0.46 km s^{-1}. That is why launch sites are placed as close to the equator as politics allows.

4.6 Questions

1. Explain, without equations, why astronauts on the space station feel weightless even though gravity there is almost as strong as on the ground. Core
2. Use Equation (4.2) to calculate the speed of a satellite in a circular orbit 1000 km above the Earth's surface. Take the Earth's radius as 6371 km. Core
3. A satellite's orbit has a periapsis 500 km above the surface and an apoapsis 7000 km above it. Find its semi-major axis and its eccentricity. Stretch
4. Use Kepler's third law and the data in Table 4.2 to estimate the period of a dwarf planet with a semi-major axis of 39.5 astronomical units. Core
5. The Moon's escape speed is 2.4 km s^{-1}. What is the speed of a circular orbit just above its surface? Why could the Apollo lunar modules leave the Moon with much smaller rockets than the ones that left the Earth? Stretch